

The Atom

Your Name

1. What is an Atom?

An atom is the smallest unit of matter that retains the properties of an element. Everything around us is made of atoms.

Atoms are extremely small and cannot be seen with the naked eye.

2. Subatomic Particles

Atoms are made of three main particles:

Particle	Charge	Mass (approx.)	Location
Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	-1	~ 0	Orbitals

- Protons and neutrons are in the nucleus.
 - Electrons move around the nucleus.
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3. Atomic Number (Z)

The atomic number (Z) is the number of protons in an atom.

- It defines the element.
 - Example: Hydrogen has $Z = 1$
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4. Mass Number (A)

The mass number (A) is the total number of protons and neutrons.

$$A = Z + N$$

Where: - Z = number of protons - N = number of neutrons

5. Electric Charge

- Neutral atom: protons = electrons
- Ion: protons $\neq$ electrons

Types of ions: - Cation: loses electrons $\Rightarrow$ positive charge - Anion: gains electrons $\Rightarrow$ negative charge

6. Isotopes

Isotopes are atoms of the same element with:

- Same number of protons (Z)
- Different number of neutrons (N)

Example: - Carbon-12 - Carbon-14

7. Relation with the Periodic Table

- The periodic table is ordered by atomic number (Z)
 - Each element has a unique Z
 - Elements in the same group have similar properties
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8. Summary

- Atoms are made of protons, neutrons, and electrons
 - Z = number of protons
 - A = protons + neutrons
 - Ions have charge
 - Isotopes differ in neutrons
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9. Exercise

Complete the following table:

Element	Z	A	Protons	Neutrons	Electrons
X	8	16	?	?	?
Y	?	23	11	?	11
Z	17	?	?	18	?

Questions:

1. Calculate the missing values.
 2. Identify if the atom is neutral or an ion.
 3. If it is an ion, indicate its charge.
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10. Challenge Question

An atom has: - 12 protons - 12 neutrons - 10 electrons

Answer:

- What is Z ?
- What is A ?
- Is it an ion?
- What is its charge?